

A LABEL-FREE CONSTRUCTION

B EXTERNAL EVALUATION

C CLAIM-GATED OUTPUTS

1 Frozen catalogue

- BlazEr1 parent (N = 5865) plus attached survey layers
- Freeze local derivatives, provenance, and manifest

2 Resolve active features

- Transform and robustly standardize the resolved inputs
- Use seven informative components; omit missing terms

3 Freeze branch and tiers

- Compute the active branch coordinate
- Fix 5 / 10 / 15 / 70 per cent tiers before TeVCat labels

4 Retrospective validation

- Attach 34 TeVCat matches only after the ranking is frozen
- Measure enrichment and harmonized AUROC with paired inference

5 Control selection history

- Compare X-ray-only, no-X-ray, and exact availability patterns
- Run random-shift and X-ray-matched controls

6 Test robustness and identity

- Weight, leave-one-out, HSP, and PCA audits
- Separate aggregate recovery from candidate stability

TeVCat labels
enter after ranking

7 Population interpretation

- Quantify the X-ray gateway and coverage-history contribution
- Map the low-z, high-peak branch and component asymmetry

8 Follow-up products

- List A: 243 stable X-ray-prioritized sources
- List B: 163 proxy-sensitive HSP-frontier sources
- Gate by redshift, attenuation, and score stability

9 External consistency checks

- Report-date chronology and observed-control pilot
- Treat as targetability checks, not detection probabilities

IACT reports and
observed outcomes